

4.1 Hledám koeficienty a, b, c . Dostanu SLR:

$$c = 1$$

$$a + c = 1$$

$$b + c = 1$$

$$a + b + c = 2$$

$$a - b + c = 0$$

$$-a + b + c = -1$$

,které zřejmě nemá řešení.

Matice:

$$\left(\begin{array}{ccc|c} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{array} \right)$$

$\underbrace{\hspace{10em}}_A \quad \underbrace{\hspace{1em}}_{\vec{z}}$

Hledám kolmý průmět vektoru pravoých stran do roviny generované sloupci nerozříděné matice.

$$A\vec{\tilde{x}} = \vec{x} \quad \vec{x} - \vec{z} \in \text{Im } A^\perp$$

$$A^*(A\vec{\tilde{x}} - \vec{z}) = 0$$

$$A^*A\vec{\tilde{x}} = A^*\vec{z}$$

$$A^*A = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix} \quad A^*\vec{z} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$\left(\begin{array}{ccc|c} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{array}\right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \\ 4 & -1 & 2 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 10 & 10 & 14 \\ 0 & 15 & 10 & 18 \end{array}\right) \sim$$

$$\sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 5 & 5 & 7 \\ 0 & 0 & -5 & -3 \end{array}\right) \sim \left(\begin{array}{ccc|c} -1 & 4 & 2 & 4 \\ 0 & 5 & 0 & 4 \\ 0 & 0 & 5 & 3 \end{array}\right)$$

$$5c = 3$$

$$c = 3/5$$

$$5b = 4$$

$$b = 4/5$$

$$-a + 16/5 + 6/5 = 4$$

$$a = \frac{20 + 16 + 6}{5} = \frac{2}{5}$$

$$z = \frac{2}{5}x + \frac{4}{5}y + \frac{3}{5}$$

$$\sum_i \left(z_i - \underbrace{(ax_i + by_i + c)}_{\bar{z}_i} \right)^2$$

$\vec{z} - \bar{z}$ je dychomy' vektor

chceme $\|\vec{z} - \bar{z}\|$ min., tedy min.

$$\sqrt{\sum_i (z_i - \bar{z}_i)^2}, \text{ tedy min. } \sum_i (z_i - \bar{z}_i)^2$$